Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	*(a)*	(i)	$n = \frac{\sin 50.0}{\sin 29.0}$ [= 1.58] **or** $\frac{v_{\text{air}}}{v_{\text{plastic}}} = \frac{\sin 50.0}{\sin 29.0}$ [or by implication] (1) $v = 1.90 \times 10^{8}$ [m s^{-1}] (1)	1	1		2	2	
		(ii)	39.3° Accept 39.2^{o} and 39^{o} e**cf**	1			1	1	
	(b)	(i)	n_{clad} [sin 90°] = 1.530 sin 81° Give this first mark even if 9° instead of 81° (1) $n_{\text{clad}} = 1.51$ (1)		2		2	2	
		(ii)	$\frac{1}{\cos 9°}$ **or** $\frac{1}{\sin 81°}$		1		1	1	
		(iii)	**Either** starting from $\Delta t = 7.5$ ns $t_{\text{axial}} = \frac{\Delta t}{0.0125}$ [= 600 ns] or by implication (1) $d = \frac{ct_{\text{axial}}}{n}$ used (1) d = 118 m, so 150 m is too long. [Accept d = 180 m so 150 m not too long, arising from omission of n – already penalised] (1) **or** if starting from d = 150 m, $t_{\text{axial}} = \frac{nd}{c}$ [= 765 ns] **or** t_{zigzag} = 775 ns or by implication (1) $\Delta t = 0.0125\ t_{\text{axial}}$ or equivalent used (1) $\Delta t = 9.6$ ns, (accept 10 ns) so 150 m is too long. [Accept Δt = 6.25 ns so 150 m not too long, arising from omission of n, already penalised.] (1)			3	3	3	
			Question 3 total	**2**	**4**	**3**	**9**	**9**	**0**